

Memo: Prof. Gottlieb meeting

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1 Modeling the Amenity Value of Medical Access and its Impact on Housing Markets

The DGLM paper we discussed focuses on market size and agglomeration. But rent acts as a congestion force: at some point agglomerating more and more medical services in a place may run into land constraints as more people want to move there. Furthermore, these effects on rents may have distributional implications: renters do worse than home-owners when market access improves, also else equal. *Uncertainty: are the effects of medical access on rent large enough to be first-order?*

The specific example I have in my mind is retirees choosing to locate somewhere because of its superior healthcare amenities, and pushing up rents for local residents.

Stylized logic. Households maximize utility

$$U_j = w_j - R_j + A_j,$$

where w_j is local wage, R_j the rent for housing, and A_j a composite amenity. Free mobility equalizes U_j across locations, so any exogenous rise in A_j —including an improvement in medical access—raises *demand* for the place. Either R_j rises or w_j falls until utility is equalized.

Uncertainty : Do we learn anything by breaking out medical access as a separate amenity?

Caveats.

- How to integrate this with the paper's original point that healthcare can be traded? The degree to which medical services are consumed by non-locals will influence how much an exogenous gain in medical access will show up in rents (in the extreme example of 0 trade costs, they shouldn't show up in rents at all).
- None of this document will consider the demand shock from healthcare providers themselves demanding housing.

Policy implications

- (i) **Incidence of travel voucher subsidies.** Improved market access is now partially captured by landlords.
- (ii) **Housing-supply policy as a complement.** Loosening land-use constraints by increasing supply elasticities in medical hubs preserves more patient surplus for renters.
- (iii) **Cross-Sectional Implications.** Boosting market access in small markets, where supply is more elastic, might seem more attractive once we consider effects on the housing market, because less of the gain is capitalized into rents.

2 Literature

Most urban/spatial papers take amenities as exogenous residuals to rationalize populations. A few papers discuss how amenities endogenously form in response to local demand, especially in response to changes in the income distribution of the neighborhood. Key papers include Diamond, 2016, Couture, Gaubert, Handbury, and Hurst, 2023, and Almagro and Domínguez-Iino, 2024.

There are also a few applied-micro papers that look at how various amenities are capitalized into house prices, such as

- Chay and Greenstone, 2005, on air quality (variation: Clean Air Act had different levels of bindingness depending on pre-existing air quality)
- Greenstone and Gallagher, 2008 examine hazardous waste (variation: discontinuity in selection for Superfund sites)
- Linden and Rockoff, 2008, examine impacts of crime risk on local property values. (variation: Megan’s Law)
- Black, 1999 examines school quality. (boundary discontinuity design)
- Alexander and Richards, 2023, in the JPubE, discuss the effect of rural hospital closures on housing prices .

3 Next steps to know if this is first-order and worth studying

1. Anecdotal evidence: do people think about healthcare access when choosing where to live?
 - The popular press, e.g. the US News and World Report, often emphasizes healthcare as a key choice in where to retire.
 - When people move somewhere for healthcare reasons, do they primarily want to be nearby for emergencies, or for monthly doctor visits?

- How “local” is this amenity? Should I think of people as moving to be within a 1-hour drive of the Mayo Clinic, or wanting to be walking distance?
2. Look for variation - how to get credible estimates of the elasticity of rent to healthcare market access.

4 Theory

This simplifies to a world where residents only care about medical access and housing costs. We must assume that wages are homogeneous across space to motivate such a model - perhaps not unreasonable for a 65+ population relying on assets/social security.

The core causal chain is:

$$\Phi_j \uparrow \implies N_j \uparrow \implies Rent_j \uparrow \implies \text{Net change in } utility$$

This contrasts with models that treat population (N_j) and rents ($Rent_j$) as exogenous.

4.1 Model Components and Assumptions

We define the following variables and relationships:

- V_j : The utility level in location j . Utility depends on market access and rents.
- Φ_j : Market access in location j (e.g., medical-access quality). This is the variable experiencing an exogenous shock, resulting in a change $d(\ln \Phi_j)$, which is a proportional change in Φ_j .
- $Rent_j$: Rent in location j .
- N_j : Population in location j .
- γ : A parameter weighting the disutility from log rents relative to log medical access
- η : The elasticity of rents with respect to population, defined as $\eta = d \ln Rent_j / d \ln N_j$. This measures the percentage change in rent for a 1% change in population.
- ϵ_{mig} : The semi-elasticity of local population (N_j) with respect to the utility level V_j . The reduced-form migration equation is $d(\ln N_j) = \epsilon_{\text{mig}} dV_j$, i.e. $\epsilon_{\text{mig}} = \frac{d \ln N_j}{dV_j}$. This is a semi-elasticity because it measures the percentage change in population N_j resulting from a one-unit absolute change in the utility level V_j .

The model is based on the following key equations:

1. **Utility:** The resident's utility is given by:

$$V_j = \ln \Phi_j - \gamma \ln Rent_j \tag{1}$$

2. **Housing Market:** Rents are determined by population, with η as the elasticity of rents with respect to population:

$$\ln Rent_j = A + \eta \ln N_j \tag{2}$$

where A is a constant. This implies:

$$d(\ln Rent_j) = \eta d(\ln N_j) \quad (3)$$

3. **Migration:** The percentage change in population, $d(\ln N_j)$, in response to absolute changes in the utility level dV_j .

$$d(\ln N_j) = \epsilon_{\text{mig}} dV_j \quad (4)$$

Here, $d(\ln N_j)$ is the percentage change in population, and dV_j is the absolute change in the utility level V_j . Thus, $\epsilon_{\text{mig}} = \frac{d(\ln N_j)}{dV_j}$ is the semi-elasticity of population with respect to the utility level V_j .

We want to find the total absolute change in utility level, dV_j , from an exogenous change in local quality, $d(\ln \Phi_j)$. We start by taking the total differential of the utility function (1):

$$dV_j = d(\ln \Phi_j) - \gamma d(\ln Rent_j) \quad (5)$$

Substitute the rent determination relationship (3) into (5):

$$dV_j = d(\ln \Phi_j) - \gamma \eta d(\ln N_j) \quad (6)$$

Substitute the migration model (4) into (6). This links the change in population back to the (endogenous) change in the utility level:

$$dV_j = d(\ln \Phi_j) - \gamma \eta (\epsilon_{\text{mig}} dV_j) \quad (7)$$

This implies

$$dV_j = \frac{d(\ln \Phi_j)}{1 + \gamma \eta \epsilon_{\text{mig}}} \quad (8)$$

Quantification / Magnitudes

I realize ϵ_{mig} doesn't have a very intuitive interpretation right now. The overall logic I hope to communicate is that the importance of the proposed mechanism depends on the product of the below three parameters.

We can consider some real-world values for the parameters:

- $\eta = 1.20$: The elasticity of rents with respect to population. This corresponds to an elasticity of housing supply of $1/\eta \approx 0.83$. Similar to Boston's level in Saiz, 2010.
- $\epsilon_{\text{mig}} = 1.5$: A 0.01 unit increase in utility increases population by 1.5%.
- $\gamma = 0.10$: The relative weight on log housing costs versus log medical access in the utility function.

The product $\gamma\eta\epsilon_{\text{mig}}$ is:

$$\gamma\eta\epsilon_{\text{mig}} = 0.10 \times 1.20 \times 1.5 = 0.10 \times 1.8 = 0.18$$

So, the absolute change in the utility level is:

$$dV_j = \frac{d(\ln \Phi_j)}{1 + 0.18} = \frac{d(\ln \Phi_j)}{1.18} \approx 0.847 d(\ln \Phi_j)$$

This means that a 1% change in Φ_j , translates into a utility change of only $0.847 \times d(\ln \Phi_j)$, with the rest being captured by landlords.

Case study: Destination Medical Center

Context. In 2013, MN launched the 20-year \$5.6 billion *Destination Medical Center* (DMC), the largest economic-development project in Minnesota’s history. DMC hopes to add 30,000 jobs to Minnesota, and help make Rochester a “global medical destination center.”¹ DMC is a public-private partnership, with $\approx 90\%$ of the funding from the Mayo Clinic, and the remainder from the city (Rochester), county (Olmsted) and state.

Popular press warns the DMC hiring wave will “*exacerbate Rochester’s pre-existing housing shortage.*” “*locals fret over housing shortage, high rents and expected population boom*”

*Thousands of new workers are set to arrive over the next few years thanks to a multibillion-dollar Mayo Clinic expansion. But Rochester isn’t building nearly enough housing to keep up with current growth.*²

Note, the story here is slightly different from the mechanism that I propose. Here, the demand for housing is increasing due to an increase of employment in the medical sector, not due to people migrating there to get medical care.

¹Source

²Source

5 Notes from Feedback on Referee Report (less well-developed)

Initial broad proposal in referee report writeup: “Spillovers - more involvement by PE firms might have GE effects beyond those nearby in geography. It’s not clear the out-of-market restriction for controls fixes this. Including some simple facts about how far people travel for healthcare might alleviate my concerns here. Text analysis on large public hospitals’ earning calls, and whether they mentioned this acquisition could also be evidence against spillovers.”

Narrower Research Question: How did competitors respond to the PE acquisition of HCA?

Broader question: How do firms respond to PE involvement in their sector, in general?

Data: Earnings calls from publicly traded hospitals. I could assemble descriptives on how firms discussed this acquisition.

Below are a few of the large publicly traded hospital systems in the US. Seeking Alpha has earnings calls back to 2007 for most of these. (\$300 to purchase all of them)

- Tenet Healthcare
- Community Health Systems
- Universal Health Services
- Encompass Health

Uncertainty: I am not sure I have a strong prior on the direction of effects, or that profit maximization predicts a major response to a competitor being acquired by a PE firm.